

## how few cycles does PyBaMM's reaction-limited SEI need to predict remaining life?

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Second-life repurposing of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited cycling time. Prior PyBaMM-based work has optimised calibration [3] and generalised physics-based degradation models [4] but has not quantified the cycling budget required for reliable RUL prediction. We ask: *how many measured cycles does a physics-based degradation model need before it can predict the remaining life of a used cell?*

We calibrate PyBaMM's [1] reaction-limited SEI degradation model (Prada2013 chemistry [2], isothermal 25 °C) against the measured fade slope of a used cell using only its first  $K$  cycles, then run the calibrated model to the full measured horizon and score against the held-out cycles  $K-N$ . A single scalar (SEI exchange current density  $j_{SEI}$ ) is fit per cell by bisection on  $\log_{10}(j_{SEI})$ . Per-cell open-circuit voltage, initial state-of-health, and ohmic resistance are constrained from HPPC and OCV-SoC characterisation. We sweep  $K \in \{50, 100, 200, 400, 800\}$  across seven MFR\_A-cohort cells with clean long-trajectory records ( $N > 1000$  cycles each) that pass an upstream reaction-limited-shape filter. Calibration and hold-out sweeps are orchestrated by a Claude Code multi-agent pipeline (AGENT\_PARAM\_ID, AGENT\_VOLTARIS\_TUNING, AGENT\_SIMULATION) released with the code.

Prediction RMSE on held-out cycles drops from a median of 11.2 pp SoH at  $K = 50$  to 1.4 pp at  $K = 100$  and 0.5 pp at  $K = 800$  (Fig. 1). At  $K = 400$ , all 7/7 cells achieve  $RMSE_{test} < 3$  pp SoH: **400 vs.  $\approx 1200$  measured cycles per cell ( $3\times$  reduction,  $\approx 4$  months of test-lab time saved at 0.25 C)**. Five of seven cells already converge at  $K = 100$  with end-of-trajectory error below 0.4 pp (Fig. 2), yielding a  **$12\times$  reduction** for that sub-cohort. The remaining two exhibit a post-formation recovery transient (positive apparent slope across cycles 50–200) and require  $K \geq 400$ ; this signal itself classifies cells for correct calibration windowing.

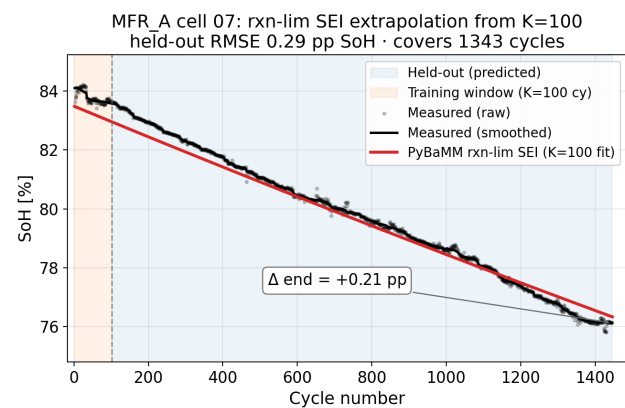
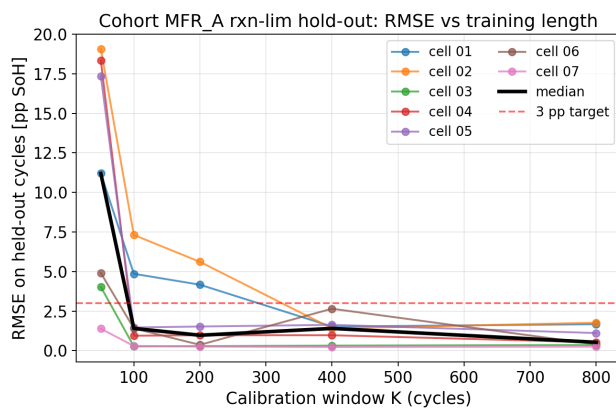


Figure 1: Held-out RMSE vs. training window  $K$ ; red dashed = 3 pp SoH target. Figure 2:  $K=100$ -cycle fit predicts remaining 1345 cycles of cell 07 to 0.29 pp SoH RMSE.

The result holds only within the reaction-limited SEI regime. Cells flagged by an upstream shape filter (non-monotonic-early, three-regime, or accelerating anomalies) are excluded — such shapes may reflect LAM domination (where PyBaMM's stress-driven LAM lacks saturation) or data-processing artefacts (batch-transition discontinuities). The full 35-run sweep is reproducible on a single core in  $\approx 30$  min using PyBaMM v26.4.3.

**References.** [1] V. Sulzer et al., *Python Battery Mathematical Modelling (PyBaMM)*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO<sub>4</sub>-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboudaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025 Book of Abstracts, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025 Book of Abstracts, p. 12.